Pre-trained model used: **ResNet50 pretrained on ImageNet**

For feature extraction, the entire ResNet50 base was frozen while the classifier was trained.

For fine-tuning, the classifier was trained first, followed by unfreezing the last 30 ResNet50 layers while keeping Batch Normalization layers frozen.

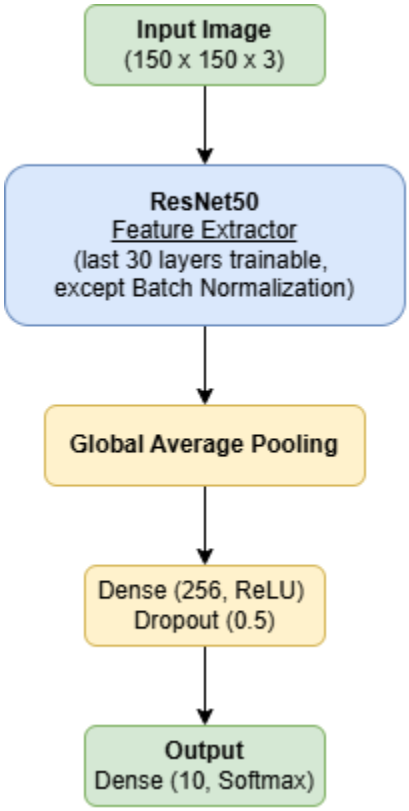


Figure - Model Architecture Diagram

Type	Accuracy	Training Time
Feature Extraction	65.1%	562.70s
Feature Extraction + Fine-tune	66.0%	1,542.51s

Table - Comparison of Feature extraction and Fine-tuning results